



Monetary policy transmission under supply chain pressure[☆]

Sebastian Laumer^a, Matthew Schaffer^{b,*}

^a University of North Carolina at Greensboro, 516 Stirling Street, Greensboro, NC 27412, USA

^b Eastern Michigan University, 703 Pray-Harrod, Ypsilanti, MI 48197, USA

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ABSTRACT

This study examines how global supply chain conditions influence the transmission of US monetary policy during the pre-pandemic period. We find that elevated supply chain pressures amplify the standard effects of monetary policy shocks on macroeconomic outcomes. For instance, peak effects on output and prices are 160 and 30 percent larger, respectively, when supply chains are stressed. This amplification arises from an intensification of the credit channel, as financial variables related to the cost of external finance become more sensitive to monetary policy under heightened supply chain pressures. For example, the peak response of the excess bond premium doubles in magnitude. Firm-level estimates further support this conclusion, with investment becoming three times more responsive to monetary policy shocks when supply chains are strained. When extending the sample beyond March 2020, the amplification effect becomes larger and occurs at longer lag lengths.

1. Introduction

Since the beginning of the COVID-19 pandemic in early 2020, the global economy has encountered unprecedented supply chain disruptions. These disruptions have been triggered by a multitude of factors, such as chip shortages,¹ the closure of ports² and production facilities, backlogs of cargo ships,³ and a lack of truck drivers⁴ and shipping containers.⁵ At the same time, expansionary fiscal and monetary policy interventions stabilized aggregate demand while households shifted their consumption away from services and towards goods. Both developments put additional pressure on already strained global supply chain systems.

The magnitude of supply chain stress can be observed in Fig. 1, which displays (Benigno et al., 2022)'s Global Supply Chain Pressure Index (GSCPI). The index measures the common component of several global and country-specific supply chain variables such as cross-border transportation costs, delivery times, backlogs, and inventory. Fig. 1 shows that supply chain pressures reached unprecedented heights during the pandemic, and, while peaking in December 2021, nevertheless remained at historically elevated levels until the end of 2022, before normalizing in 2023. Speculation surrounding deglobalization and reshoring, a re-prioritization of robustness over efficiency, environmental factors, shocks to trade routes and contentious geopolitical developments suggest that

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* Corresponding author.

E-mail addresses: s_laumer@uncg.edu (S. Laumer), mschaff5@emich.edu (M. Schaffer).

¹ <https://www.latimes.com/business/story/2022-07-01/chip-shortage-keeps-driving-up-auto-prices>.

² <https://www.cnn.com/2022/05/06/business/china-lockdowns-global-port-chaos-supply-chains-intl-hnk/index.html>.

³ <https://www.cnn.com/2021/10/20/business/la-long-beach-port-congestion-problem-national-impact/index.html>.

⁴ <https://www.nbc26.com/news/local-news/expect-to-pay-more-at-the-pump-increased-demand-for-gas-due-to-a-national-truck-driver-shortage>.

⁵ https://www.joc.com/article/california-ports-look-inland-short-term-container-storage_20220407.html.

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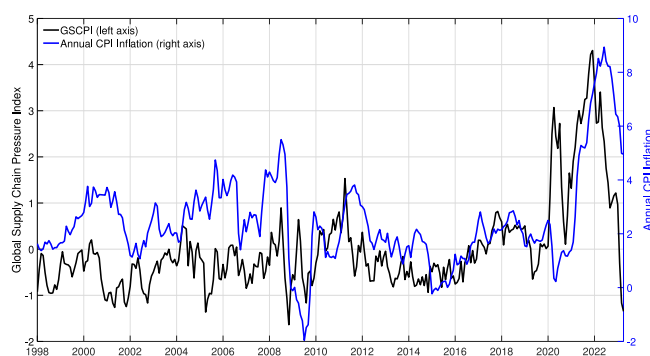


Fig. 1. Global Supply Chain Pressure Index. Note: The figure shows the Global Supply Chain Pressure Index (GSCPI) developed by Benigno et al. (2022) in black as well as annual rates of CPI inflation in blue. Although global supply chain pressures have reached unprecedented levels since the onset of the pandemic, there have also been notable spikes prior to 2020, e.g., in 2008 during the Great Financial Crisis, in 2011 after the Great East Japan Earthquake, or in 2017/18 following the U.S.-China trade war. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

global supply chain pressures may remain volatile for years to come. As a result, understanding the macroeconomic implications of supply chain disturbances takes on a new urgency.

The most recent, pressing challenge for policymakers has been combating the decades-high rates of inflation since 2021, also shown in Fig. 1. Central banks around the world have sharply increased interest rates in an effort to reduce inflation. For example, in 2022 alone, the Fed raised the federal funds rate by 425 basis points including several rounds of 75 basis points hikes. In Spring 2024, the federal funds rate is at its highest level since the early 2000s. Other central banks such as the Bank of England and the European Central Bank have followed the Fed's example. The economic environment following the COVID-19 pandemic has therefore been characterized by unprecedented levels of supply chain pressure, decades-high rates of inflation and an aggressive tightening of monetary policy. These circumstances motivate the purpose of our study – to examine how supply chain disruptions influence the transmission of monetary policy.

This paper applies (Jordà, 2005)'s nonlinear local projection (LP) method and Benigno et al. (2022)'s GSCPI to study how the impact of monetary policy shocks on macroeconomic outcomes varies with the underlying state of supply chain conditions. We uncover that the effects of monetary policy are significantly more pronounced when supply chains face greater disruption. For instance, a contractionary monetary policy shock lowers output and consumer prices in any state, but the decline is more than twice as large for output, and 30% larger for consumer prices, when supply chain pressures are elevated. We observe similar amplification effects for other key macroeconomic variables such as retail sales and the unemployment rate. Importantly, these estimated amplification effects are robust even after considering various potential confounding factors including the business cycle, financial conditions, energy and oil prices, as well as geopolitical events.

We show that the amplified response of macroeconomic outcomes during times of supply chain stress can be explained by an intensification of the credit channel of monetary policy. As argued by, e.g., Bernanke and Gertler (1995), in the presence of financial frictions monetary policy will affect borrowing costs not only through the general level of interest rates, but also through an external finance premium which reflects the fundamentals of a borrower's financial position. We argue that borrower fundamentals become more sensitive to changes in the expected path of interest rates when supply chains face greater disruption. Indeed, in our nonlinear LP framework we show that financial variables related to the cost of external finance become more sensitive to monetary policy when supply chains are strained. For instance, following a contractionary monetary policy shock, stock prices decrease 3–4 percentage points more while the excess bond premium increases by an additional 0.13 basis points when supply chain pressures are high. With the external finance premium becoming more sensitive, monetary policy has a larger impact on borrowing costs, which, in turn, results in an amplified effect on investment and aggregate demand, and hence, on output and inflation.

Our mechanism proposes that firms face higher interest rate increases following a contractionary monetary policy shock when supply chains are disrupted. As a result, firms are expected to lower their investment more strongly. We provide additional evidence in line with this argument using micro data from Compustat. Consistent with our baseline results, we find that firms lower their investment significantly more i.e., three times as much, following a contractionary monetary policy shock when supply chain pressures are elevated. Notably, while investment becomes more responsive to monetary policy for all firms, the increased responsiveness in the tight supply chain state is largest for firms that face relatively greater financial constraints. The larger differential response for firms that are most affected by financial frictions provides further support for our hypothesis that supply chain disruptions amplify the macroeconomic impact of monetary policy through an intensification of the credit channel.

As discussed by Ng (2021), many economic time series may have experienced structural breaks in their data generating processes during the pandemic. The pandemic period also introduces a host of factors, like government restrictions on mobility and business openings, that are likely to confound empirical analysis. Accordingly, we restrict our main focus to a pre-pandemic sample from January 1998 to December 2019. However, when extending the sample through March 2023, we continue to observe an amplified impact of monetary shocks on macroeconomic outcomes which is considerably stronger compared to the estimated amplification effect for the pre-pandemic sample. Most importantly, there is a longer lag time to the trough/peak of the responses. The longer lag

time until peak effect is consistent with the persistence of inflation and the resilience of real activity throughout the Fed's aggressive 2022 tightening campaign.

Our results have vital implications for the current policy environment. They suggest that the expansionary policies enacted by the Fed at the onset of the pandemic may have stimulated demand and contributed to inflationary pressures more strongly than expected due to the unprecedented degree of supply chain stress. Furthermore, the longer estimated monetary policy lags from the extended sample as well as improved supply chain conditions observed since 2023 suggest that the Fed's "last mile" in combating inflation may be the most challenging.⁶ More generally, our results highlight the need for central bank authorities to monitor and consider the state of supply chain conditions when formulating monetary policy.

Our study is related to several strands of the literature. First, our study relates to the literature on monetary policy and supply chains. Theoretical studies such as Carvalho et al. (2021a), Pasten et al. (2020), and Nakamura and Steinsson (2010) analyse the role of production networks in enhancing strategic complementarities in price setting and thus increasing short-run money non-neutrality. Ghassibe (2021) provides empirical evidence in favour of this view, finding that the amplification effect of input–output linkages can account for at least 30% of the overall effect of monetary shocks on aggregate consumption. Ozdagli and Weber (2021) find that at least half of the effect of monetary shocks on stock returns can be attributed to indirect production network linkages. The literature has, therefore, identified important ways in which the *existence* of input–output linkages in a production network can amplify the effects of monetary policy.

In addition, Ascari et al. (2024) build a two-country NK model in which the optimal response of monetary policy to supply chain disruptions depends on the country's participation in global value chains. Balleer and Noeller (2023) use German survey data and show that material-constrained firms increase prices significantly more often in response to monetary policy compared to unconstrained firms. Comin et al. (2024) demonstrate that supply chain disruptions "set the stage" for monetary policy. The authors show that in 2021, disruptions in the supply chain, coupled with accommodative monetary policy, resulted in binding supply constraints and an acceleration of inflation. In contrast, during the second half of 2022, monetary tightening alleviated supply constraints, leading to a reduction in inflation. Bai et al. (2024) also study how the impact of monetary policy depends on supply chain disruptions using a threshold VAR model for the period between January 2017 and September 2023. In line with our results, inflation responds more strongly when supply chains are stressed. We build upon this pre-existing literature by analysing how the effect of monetary policy on macroeconomic aggregates varies with the *condition* of global supply chains. Using time series data in a flexible LP framework, we provide novel evidence that elevated supply chain pressure amplifies the effects of monetary policy on macroeconomic variables.

Second, our paper connects to the literature on production networks and input–output linkages as an amplification mechanism for economic shocks, see e.g., Acemoglu et al. (2016), Barrot and Sauvagnat (2016), Boehm et al. (2019), Acemoglu and Tahbaz-Salehi (2020) and Carvalho et al. (2021b). Empirically, Barrot and Sauvagnat (2016) and Boehm et al. (2019) exploit natural disasters as exogenous shocks. They show that the shocks not only affected firms that were directly hit by the natural disasters, but also the output and sales of their suppliers and customers. Carvalho et al. (2021b) estimate the macroeconomic effects of the 2011 Great East Japan Earthquake. They find that the decline in Japan's real GDP growth following the earthquake would have been significantly less severe in the absence of input–output linkages. We contribute to this literature by showing that input–output *disruptions* amplify the effect of monetary policy shocks.

Third, this study also relates to the credit channel literature that links monetary policy and financial frictions, see e.g., Bernanke and Gertler (1995), Bernanke et al. (1999), Gertler and Karadi (2015), and Ottonello and Winberry (2020). While Smirnyagin and Tsyvinski (2022) report that equity premia increase when individual firms are faced with idiosyncratic supply chain disruptions, we provide novel evidence that the cost of external finance becomes more sensitive to monetary policy in the presence of aggregate supply chain disturbances. Specifically, we show that contractionary monetary policy shocks raise (lower) credit spreads and term premia (stock prices) significantly *more* when supply chain pressures are high, thereby providing an economic explanation for the larger effect of monetary policy on macroeconomic aggregates.

Fourth, our analysis relates to the literature on the nonlinear effects of monetary policy, see e.g., Weise (1999), Garcia and Schaller (2002), Peersman and Smets (2002, 2005) and Tenreyro and Thwaites (2016). These papers analyse how the effect of monetary policy varies with the business cycle. While the earlier studies report a stronger effect of monetary policy during recessions, Tenreyro and Thwaites (2016) find that the effects of monetary policy are "less powerful" during recessions. Our study differs from the pre-existing literature by analysing the state-dependency of monetary policy with respect to supply chain conditions. Importantly, the GSCPI is purged of demand-side factors related to business cycle fluctuations, which ensures that our results are not driven by co-movement with the business cycle.⁷ We show that the macroeconomic effect of monetary policy also varies significantly with the degree of supply chain pressure.

Lastly, our paper is related to the small but growing literature on aggregate supply chain shocks, e.g., Finck and Tillmann (2022), Kuehl et al. (2022), Burriel et al. (2023), Gordon and Clark (2023), Bernanke and Blanchard (2023), Ascari et al. (2024), Bai et al. (2024) and Elsayed et al. (2023). The findings of this literature suggests that shocks to (global) supply chains lead to increases in consumer, producer and import prices. The focus of this study differs from the aforementioned series of papers. We do not study the effects of supply chain shocks, but instead focus on how the effects of US monetary policy shocks are influenced by the state of global supply chain disruptions.

The paper proceeds as follow. Section 2 explains our methodology. Section 3 presents our main results. Section 4 discusses firm-level evidence. Section 5 extends our baseline analysis through the COVID-19 pandemic. Section 6 discusses and concludes.

⁶ See e.g., also <https://www.reuters.com/markets/us/feds-powell-jefferson-square-restrictive-policy-with-strong-data-2024-04-16/>.

⁷ According to Benigno et al. (2022), demand-side factors may contaminate the GSCPI to some degree around the Global Financial Crisis, however our baseline results hold even when explicitly controlling for this period.

2. Methodology

This section describes a number of important details of our econometric methodology. We first generally explain Jordà (2005)'s local projection method that we use to estimate impulse response functions. We then adapt the method to our nonlinear setting. Finally, we outline the Global Supply Chain Pressure Index (GSCPI), developed by Benigno et al. (2022), which is used to capture the degree of supply chain pressure in the economy.

2.1. Local projection method

This study uses (Jordà, 2005)'s local projection method to estimate impulse response functions. Following Coibion et al. (2017), we can express local projections as

$$\ln(x_{t+h}) = c^h + \sum_{j=1}^J \alpha_j^h \ln(x_{t-j}) + \sum_{j=0}^J \beta_j^h \text{shock}_{t-j} + \sum_{j=0}^J \Gamma_j^h Z_{t-j} + \epsilon_{t+h}, \quad h = 0, \dots, H, \quad (1)$$

where c is a constant, x is the outcome variable of interest, shock is the macroeconomic shock of interest, Z is a vector of control variables, and ϵ is the residual. The local projection method involves a series of regressions for each variable of interest for each of the H impulse response horizons. The key object of interest is the parameter β_0^h . This parameter captures the effect of the macroeconomic shock on the variable of interest h periods after the shock hits the economy. The series of parameters $\{\beta_0^h\}_{h=0}^H$ then gives the impulse response function, i.e., the response of the variable of interest to the shock over time.

Here, we are interested in the macroeconomic effects of a monetary policy shock. To identify a monetary policy shock, we follow the most recent advancements in the literature and rely on high-frequency identification, e.g., following Gürkaynak et al. (2005), Gertler and Karadi (2015) and Nakamura and Steinsson (2018). The idea behind this approach is to isolate the change in a financial variable(s) that captures the stance of monetary policy in a tight window around a central bank policy announcement. As long as no other market moving events occur within the window, the change in the price of the financial variable(s) around the announcement will capture any unanticipated changes in monetary policy. The resulting time series of “surprises” can then be used as a measure of the monetary policy shock.

The monetary shock used in our baseline specification is the daily change in the two-year Treasury zero-coupon yield around FOMC announcements.⁸ We focus on changes in the two-year nominal yield for two reasons. First, as argued by Hanson and Stein (2015), it allows us to capture revisions to the expected path of interest rates over a two-year horizon in a straightforward manner – in other words, it accounts for surprise changes in forward guidance in addition to immediate federal funds rate surprises. This is especially important during the zero lower bound (ZLB) period. Second, it allows us to maintain a consistent shock measure across all sample periods, which becomes important when extending our sample in Section 5.

We use two additional monetary shock measures for robustness checks. The first is akin to the “policy news shock” from Nakamura and Steinsson (2018). Specifically, it is constructed using four Eurodollar futures contracts, expiring 1 quarter ahead (ED1) to 4 quarters ahead (ED4). We perform a principal component analysis on the daily changes in the four futures prices around FOMC meetings. The first principal component, which accounts for over 90% of the variation across the four contracts, is then used as the alternative shock measure.⁹ We also use the unified monetary shock measure from Bu et al. (2021), which has the advantage of being uncontaminated with potential central bank information effects, as an additional robustness check. For all three measures, a monthly shock series is constructed by summing all high frequency shocks occurring in a given month and setting the series to zero in months without a FOMC meeting. Each measure is normalized so that the impulse response coefficients can be interpreted as the effect of a one standard deviation shock.

Local projections are easily adaptable to nonlinear or state-dependent settings like the one underlying our research question, see e.g., Jordà (2005) or Ramey and Zubairy (2018). For example, Eq. (1) can be extended to

$$\begin{aligned} \ln(x_{t+h}) = c^h &+ \sum_{j=1}^J \alpha_j^h \ln(x_{t-j}) + \sum_{j=0}^J \beta_j^h mp_{t-j} + \psi_j^h GSCPI_{t-1} \\ &+ \sum_{j=0}^J \delta_j^h GSCPI_{t-1} mp_{t-j} + \sum_{j=0}^J \Gamma_j^h Z_{t-j} + \epsilon_{t+h}, \quad h = 0, \dots, H, \end{aligned} \quad (2)$$

where $GSCPI$ is the Global Supply Chain Pressure Index (GSCPI) which is described in detail in the next subsection, and mp is the high-frequency monetary policy shock. In a nutshell, the GSCPI captures the state of global supply chains and is used here to distinguish between different states of aggregate supply chain conditions. In Eq. (2), the main objects of interest are β_0^h and δ_0^h . The GSCPI is normalized so that β_0^h measures the effect of a monetary policy shock on the variable of interest when global supply chain pressures are at their sample average, i.e., when supply chain conditions are “normal”. By contrast, δ_0^h captures the *additional* impact of a monetary policy shock when global supply chain pressures are high, i.e., one standard deviation above the sample average. In Section 3, we use $\{\beta_0^h\}_{h=0}^H$ and $\{\beta_0^h + \delta_0^h\}_{h=0}^H$ to study the impulse responses of key economic variables to a monetary policy shock when supply chain pressures are normal and elevated, respectively. Moreover, we use $\{\delta_0^h\}_{h=0}^H$ to test whether the difference in

⁸ Zero-coupon yields are from Gürkaynak et al. (2007).

⁹ Our Eurodollar futures data ends in 2019, hence we are not able to use this shock when extending the sample period to include the pandemic.

Table 1
Summary statistics (Jan. 1998–Dec. 2019).

	Mean	Median	Std Dev	Min	Max	Obsv.
Shock						
Monetary policy shock	−0.04	0.00	1.00	−4.17	7.82	264
State variable						
Global supply chain pressure index	0.00	−0.09	1.00	−2.59	3.70	264
Macro variables						
Industrial production	4.56	4.57	0.06	4.43	4.65	264
Consumer price index	5.34	5.37	0.14	5.09	5.55	264
Unemployment rate	5.74	5.10	1.78	3.50	10.00	264
Retail sales	12.80	12.80	0.22	12.35	13.17	264
Financial variables						
S&P 500	7.28	7.19	0.34	6.60	8.08	264
Excess bond premium	0.11	−0.11	0.71	−1.09	3.47	264
Two-year term premium	0.17	0.17	0.27	−0.35	0.83	264
Ten-year term premium	0.56	0.55	0.56	−0.74	1.85	264

Note: The table presents summary statistics for variables used in the empirical analysis. Data is monthly over a January 1998 to December 2019 sample period. IP, CPI, retail sales, and the S&P 500 are in log form.

responses across supply chain states is statistically significant. AIC values suggest setting $j = 2$, hence we include two lags of all right hand side variables in our estimation.

Local projections have become a popular alternative to SVAR models. The most important advantage of local projections over SVAR models in our context is their easy adaption to nonlinear settings. Nonlinear local projections are estimable via least squares, while nonlinear SVAR models rely on numerical methods and/or complicated Bayesian estimation techniques, see e.g., [Auerbach and Gorodnichenko \(2012\)](#) or [Laumer and Philipps \(2023\)](#). Furthermore, nonlinear local projections can be estimated without making assumptions about how the economy transitions from one state to another after it is hit by an shock. The regime changes are reflected by the estimated parameters, see e.g., [Auerbach et al. \(2013\)](#), [Tenreiro and Thwaites \(2016\)](#) or [Ramey \(2016\)](#). By contrast, in nonlinear SVAR models, the user must make assumptions on how impulse response functions are computed, see e.g., [Auerbach and Gorodnichenko \(2012\)](#), [Caggiano et al. \(2015\)](#) and [Laumer and Philipps \(2023\)](#). Moreover, we can estimate the responses for any variable of interest via local projections, while we are constrained by the curse of dimensionality in (nonlinear) SVAR models. This is particular important when we explore potential transmission channels for our main results. Finally, the number of parameters in SVAR models are relatively high compared to LPs, especially in nonlinear settings. Given that our state variable only becomes available in the late 1990s, LPs are more appropriate in our context. A similar argument can be found in [Auerbach et al. \(2013\)](#).

There are various ways to implement nonlinear local projections. For instance, [Tenreiro and Thwaites \(2016\)](#) apply smooth-transition local projections, while [Ramey and Zubairy \(2018\)](#) consider a threshold version. We prefer the flexible specification in Eq. (2) because it allows us to transparently estimate whether the two sets of impulse response functions are statistically different.¹⁰ Moreover, the threshold and smooth-transition LPs require the estimation of two full sets of parameters, one set for each state, while our specification only requires us to estimate an additional interaction term. Given that our sample begins in the late 1990s, this is an important advantage.

Summary statistics for the variables of interest are presented in [Table 1](#). There are 264 monthly observations from January 1998 through December 2019. All macro variables other than the unemployment rate are in log form. For the financial variables, all variables are in percentage points, except the S&P 500 which is in log form. The monetary policy shock is normalized so that a unit change is equivalent to a one standard deviation change. The GSCPI is normalized to have mean zero and unit standard deviation over the 1998–2019 sample period.

2.2. Global supply chain pressure index

The nonlinear version of the local projection method requires a variable that captures the underlying state of the economy that the researcher wants to analyse. In this study, we are interested in how the effect of a monetary policy shock may or may not depend on global supply chain conditions. Consequently, we need to employ a variable that captures the state of supply chain pressures. [Benigno et al. \(2022\)](#) from the Federal Reserve Bank of New York have developed such a variable, the Global Supply Chain Pressure Index (GSCPI).

To construct the GSCPI, [Benigno et al. \(2022\)](#) collect data on 27 variables. The first six measure cross-border transportation costs: the Baltic Dry Index, which tracks the cost of shipping raw materials; the Harper Index, which tracks container shipping rate changes; and US Bureau of Labor Statistics price indices, which track the cost of air transportation of freight to and from the United States from Asia and Europe. The remaining variables capture country-level manufacturing conditions from China, the European Union, Japan, South Korea, Taiwan, the UK and the US from Purchase Manager Index (PMI) surveys. For each country, data on manufacturing delivery times, backlogs, and purchased stocks is incorporated.

¹⁰ Recall that if δ is statistically significant, then the two sets of impulse response differ statistically.

Table 2
Periods with GSCPI > 1.

Dates	GSCPI	Notable events
April–June 2004	1.45–1.53	Middle east tension
February 2008	1.36	Great recession
June–August 2008	1.05–2.55	Great recession
April 2009	1.92	Great recession
March–May 2010	1.09–1.38	Iceland volcano eruption
August 2010–May 2011	1.33–3.70	Tohoku earthquake
July 2011	1.10	Tohoku earthquake
January 2012	1.23	Thailand floods
February 2017	1.02	US-China trade dispute
August 2017–January 2018	1.46–2.31	US-China trade dispute
March 2018–January 2019	1.30–1.70	US-China trade dispute

Note: The table lists periods in which global supply chain pressures were more than one standard deviation above the 1998–2019 average.

The authors purge the 27 variables of demand side factors. They regress the six cross-border transport costs variables individually on GDP-weighted averages of the “new orders” and “quantities purchased” PMI subcomponents, where “new orders” proxy for customer demand for firms’ products and “quantities purchased” proxies for producers’ input demand. The residuals, which can be interpreted as the variation in the transport cost measures attributable to supply factors, are then used as inputs for the GSCPI. Similarly, the 21 country-specific measures on manufacturing conditions are individually regressed on the contemporaneous value and two lags of that country’s “new orders” subcomponent and the residuals are taken as inputs to the GSCPI. After purging the 27 variables of demand-side influence, Benigno et al. (2022) perform a principal component analysis and take the first principal component as the Global Supply Chain Pressures Index.

The GSCPI, as displayed in Fig. 1, is a monthly index beginning in January 1998. It is normalized so that a value of zero means supply chain pressures are at the sample average.¹¹ A positive value represents how many standard deviations the index is above average, i.e., represents elevated supply chain pressures. A negative value corresponds to eased supply chain pressure. Apart from a few periods, the GSCPI is negative for the early part of the sample. Hence, supply chain pressures were below average between 1998 and 2008. During the Global Financial Crisis, the GSCPI bounces up and down. In 2011, the GSCPI reaches its pre-COVID-19 peak due to a flooding in Thailand and the Tuhoku earthquake in Japan, which led to substantial supply chain disruptions in manufacturing even in the US, see e.g., Barrot and Sauvagnat (2016). Between 2012 and 2017, the GSCPI is negative before it begins to rise in 2017 due to the tariff battle between the US and China (Benigno et al., 2022). Since the outbreak of the COVID-19 pandemic the GSCPI has spiked to previously unseen levels, reaching a peak in late 2021. The GSCPI remained elevated throughout 2022, before declining rapidly in 2023. Table 2 summarizes periods with elevated supply chain pressure (GSPCI values exceeding 1) over our pre-pandemic sample.

Fig. 2 compares the GSCPI with other measures of supply chain disruptions. The top-left panel compares the GSCPI to Elsayed et al. (2023)’s Global Sectoral Supply Disruption Index (GSSDI). The two indices capture similar developments over time, e.g., the Great Recession and the Covid-19 pandemic, and are highly positively correlated (coefficient: 0.67). In the top-right panel, we compare the GSCPI to the share of disrupted trade pairs from Smirnyagin and Tsyvinski (2022). The measures are positively correlated (0.39) and display spikes at similar times, e.g., during the Great Recession and following the Tohoku Earthquake in Japan. In the bottom-left panel, we consider (Soto, 2023)’s Supply Chain Bottleneck Index (SCBI). Both capture similar dynamics and are highly correlated (0.70). Lastly, the bottom-right panel shows the GSCPI along with the Bloomberg Supply Constraint Indicator.¹² The two indexes are also positively correlated (0.53) and exhibit similar behaviour around key dates, e.g., 2011 Earthquake in Japan, during the US-China tariff dispute and the COVID-19 pandemic. Section 3.3 considers the GSSDI and SCBI to demonstrate the robustness of our results. By contrast, the other two measures only become available in 2008, making them ill-suited to be used in our empirical analysis. However, the fact that all are highly positively correlated with the GSCPI, and share similar dynamics around key events, supports our use of the GSCPI as a gauge of aggregate supply chain conditions.

Lastly, Appendix Figure A.2 compares the GSCPI with our preferred monetary policy shock series. The figure shows that there is no systematic relationship between the GSCPI and the monetary policy shock, i.e., monetary policy shocks are not larger or more likely to be positive when the GSCPI is elevated.¹³

3. Results

This section presents our main results. We first show the impulse responses of macroeconomic aggregates and then the impulse responses of financial variables related to the cost of external finance. Section 3.3 provides additional exercises and robustness

¹¹ The GSCPI as displayed in Fig. 1 is normalized relative to a sample ending in 2023. In our empirical analysis, we re-normalize relative to our 1998–2019 sample period, so that a value of zero represents average pre-pandemic conditions. The re-normalized GSCPI is plotted in Appendix Figure A.1.

¹² <https://www.bloomberg.com/toaster/v2/charts/e0919ea0c2234156901aaafa49203249.html?brand=business&webTheme=light&web=true&hideTitles=true>.

¹³ The correlation between January 1998 and December 2019 is 0.0601.

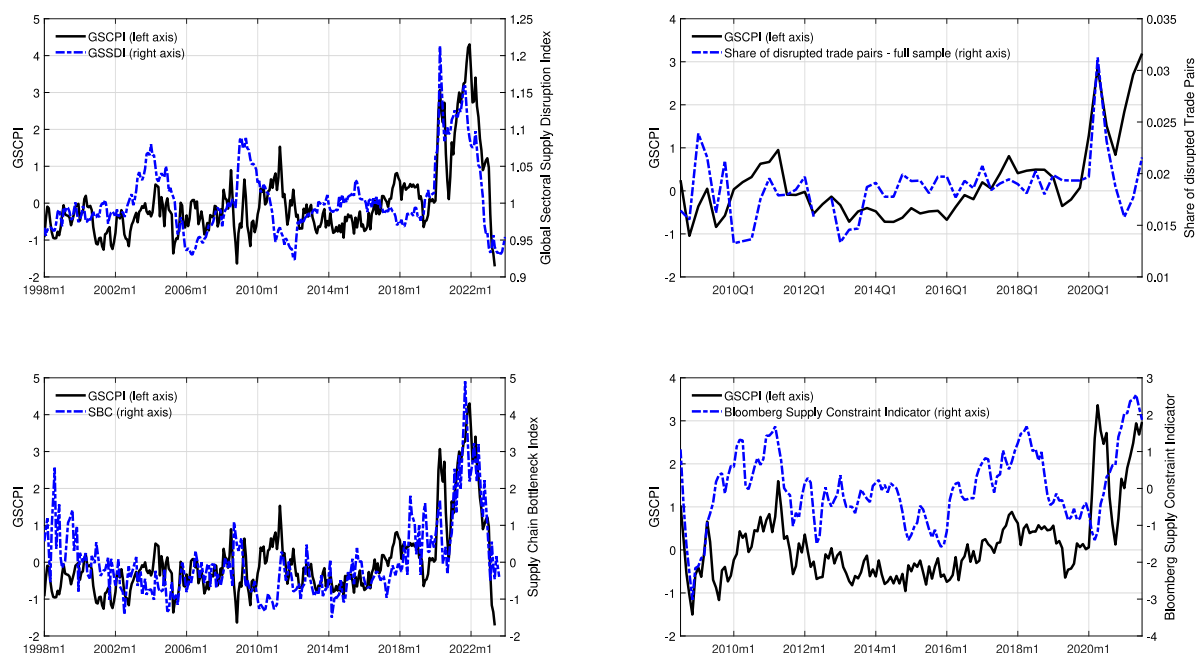


Fig. 2. Comparison of the GSCPI with other measures of Supply Chain Disruption. Note: The figure compares the Global Supply Chain Pressure Index (GSCPI) from Benigno et al. (2022) with alternative measures of supply chain disruptions. The upper-left panel considers (Elsayed et al., 2023)'s Global Sectoral Supply Disruption Index (GSSDI). The upper-right subplot includes the share of disrupted trade pairs from Smirnyagin and Tsyvinski (2022) over their quarterly 2008 to 2021 sample. The lower-left panel compares the GSCPI to Soto (2023)'s Supply Chain Bottleneck Index (SCBI), and lastly, the lower-right subplot includes the Bloomberg Supply Constraint Indicator over a monthly 2008 to 2021 sample.

checks. All responses are normalized to a one standard deviation contractionary monetary policy shock. Point estimates are illustrated by the blue solid line, 68-percent confidence intervals are in dark grey and 90-percent confidence intervals are in light grey with dashed lines.

3.1. Macro variables

Fig. 3 shows the impulse responses of industrial production, the consumer price index, the unemployment rate and retail sales to a contractionary monetary policy shock when supply chain conditions are average (left subplots) and tight (right subplots).¹⁴

Our baseline results show that heightened supply chain pressures amplify the effects of a monetary policy shock on aggregate macroeconomic outcomes. In the top left panel, a contractionary monetary policy shock leads to a decline in industrial production with a short delay in both supply chain states.¹⁵ However, the decline is stronger and longer lasting when supply chains are disrupted. Specifically, there is a statistically significant 0.20% decline in IP six months after the shock when supply chain conditions are average. By contrast, when supply chain pressures are one standard deviation above average, the peak effect is more than double in magnitude – a 0.54% decline, fifteen months after the shock. Not only is the peak effect larger in magnitude when supply chain pressures are elevated, but the decline is also significant for a longer period – from roughly three to twenty months following the shock.

Fig. 3 depicts a similar result for CPI and retail sales. A contractionary shock reduces consumer prices and retail sales by more and for a longer duration when supply chains face greater disruption. The decline in CPI reaches a statistically significant trough after eight months in either state, but the magnitude is nearly 30% larger when supply chain pressures are elevated (−0.21%) compared to when pressures are at their historical average (−0.16%). The amplified inflation response is consistent with (Comin et al., 2024), who show that the interaction of monetary policy shocks with supply chain disruptions explains a sizable portion of the path of inflation in 2021–2022. For retail sales, the peak decline is over twice as large (−0.75% vs. −0.32%) when supply chains are disrupted, and the duration of the decline is much longer, lasting from six to twenty-four months after the shock.

Lastly, the unemployment rate increases by a much larger magnitude and for a longer duration following a contractionary shock when supply chain pressures are greater. Indeed, the peak effect when supply chain conditions are average is a marginally significant 0.04 percentage point increase ten months after the shock. In contrast, the peak effect under elevated pressures is a 0.22 percentage

¹⁴ In all figures, “tight supply chain conditions” refers to a state where the GSCPI is one standard deviation above average.

¹⁵ Our figures display impulse responses to a positive (contractionary) shock, but it should be noted that the estimates are symmetric, i.e., flipping the sign gives the response to a negative (expansionary) shock.

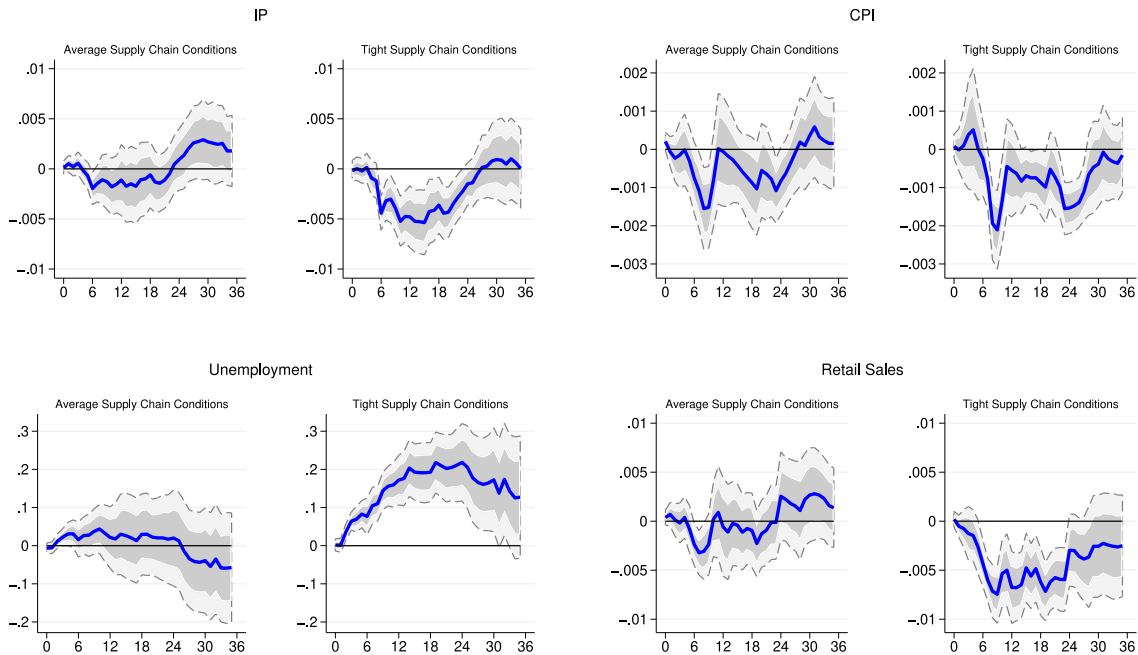


Fig. 3. State-dependent impulse responses. Macro variables. Note: The figure shows the state-dependent impulse response of industrial production, the consumer price index, the unemployment rate and retail sales to a contractionary monetary policy shock. The left panel shows the response (β_0^h) when supply chain pressures are at their sample (1998–2019) average. The right panel shows the response when supply chain pressures are one standard deviation above average ($\beta_0^h + \delta_0^h$). The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

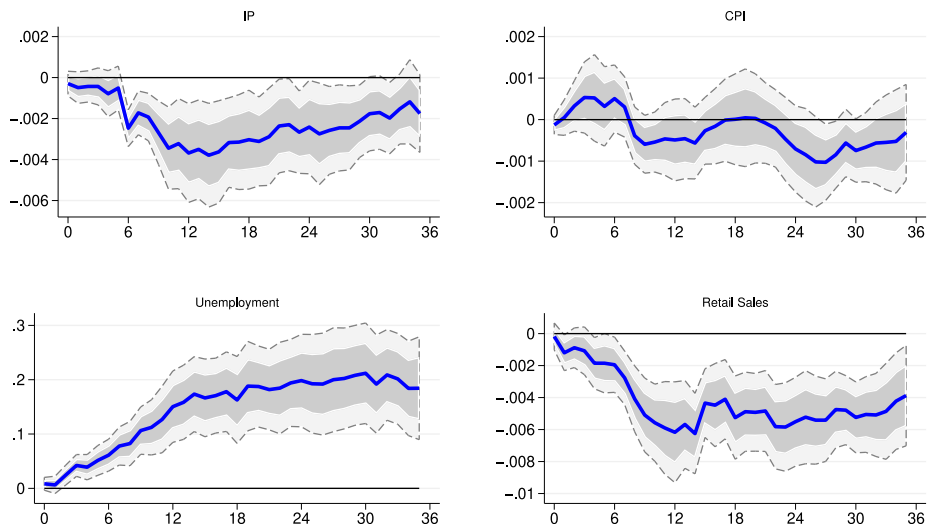


Fig. 4. Differential responses. Macro variables. Note: The figure shows the *additional response* (δ_0^h) of industrial production, the consumer price index, the unemployment rate and retail sales to a contractionary monetary policy shock during a period of elevated supply chain pressure, relative to an average period. The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

point increase after twenty-four months, and the increase remains statistically significant over a twenty-seven month longer horizon (from three to thirty months after the shock).

Fig. 4 illustrates that the responses across states are also statistically different, i.e., the figure shows that the *additional effect* of a contractionary monetary policy shock when supply chains are disrupted, δ_0^h , is statistically different from zero. With a short delay of a few months, we find this to be the case for all four macro variables. The responses of industrial production, retail sales and the unemployment rate remain statistically different for at least twenty-four months. The difference of the CPI response is more nuanced

and only statistically different at the 68-percent confidence interval between seven and fourteen months as well as between twenty-four and thirty-four months after the shock. Overall, our results indicate that the standard responses of macroeconomic outcomes to a monetary shock, e.g., as shown in Gertler and Karadi (2015), are significantly larger in the face of heightened supply chain pressures.

3.2. Financial variables

The question that naturally follows is, what explains the larger impact of monetary policy shocks in states of heightened supply chain pressure? In this study, we argue that this amplification effect occurs due to an intensification of the credit channel of monetary transmission. Consider a simple illustration of the credit channel from Bean et al. (2002). The interest rate faced by firms, R_t , is equal to the risk-free rate plus an external finance premium:

$$R_t = R_t^* + f_t(D_t/E_t) \quad (f' > 0) \quad (3)$$

where the risk-free rate, R_t^* , is the central bank's policy tool and the external finance premium is a function of the firm's financial position, represented here as a simple debt-to-equity ratio. A long line of literature, including (Bernanke and Gertler, 1995), has argued that changes to the risk-free rate will directly and indirectly alter a firm's financial position. Since many firms finance inventories and working capital with short-term debt, an increase in interest rates will directly increase borrowing costs ($\uparrow D$). To the extent that higher rates depress demand for a firm's product, there will also be a decline in the expected future cash flows that determine a firm's equity value ($\downarrow E$). A weaker balance sheet, as stylized through a higher debt-to-equity ratio ($\uparrow D/E$), increases the external finance premium ($\uparrow f(D/E)$) and makes it even more costly for firms to access external funds. A policy-induced increase in the risk-free rate will, therefore, increase firm borrowing costs more than one-for-one due to the rise in the external finance premium.

The credit channel of monetary transmission is operative in any state of the world that exhibits information asymmetries, or frictions, in financial markets. However, we postulate that the external finance premium becomes *more* sensitive to changes in the risk-free rate in a state of heightened supply chain pressure. This heightened sensitivity can arise from multiple sources. When supply chains are disrupted, and there is greater uncertainty around the availability of upstream supply or the timing of downstream demand, firms may desire to keep larger stocks of inventories or rely more heavily on working capital. Accordingly, with larger short-term debt requirements, a given increase in interest rates will result in a larger increase in borrowing costs ($\uparrow\uparrow D$). Alternatively, production delays resulting from input scarcity or revenue uncertainty stemming from downstream disruptions, may produce larger downward revisions to expected future cash flows ($\downarrow\downarrow E$) following an increase in rates. It is also possible that investors and lenders place a greater weight on a firm's financial position during times of aggregate supply chain distress ($f'_{\text{tight}} > f'_{\text{average}}$). In either case, $f(D/E)$, and hence, R_t increase more strongly following an increase in R_t^* when supply chains are disrupted.

Regardless of the precise source, if a strengthening of the credit channel underpins the amplified response of macroeconomic variables documented in Section 3.1, we would expect to observe an increased sensitivity of external financing costs to monetary policy when global supply chains are disrupted. We use the nonlinear LP framework from Eq. (2) to examine this hypothesis. Specifically, we replace the four macroeconomic outcome variables with four financial variables related to the cost of obtaining external finance: the S&P 500 stock index, the (Gilchrist and Zakrajšek, 2012) excess bond premium, and the two-year and ten-year Treasury term premia.¹⁶ The contemporaneous values and two lags of IP, CPI, the unemployment rate, and retail sales are included as controls on the right-hand side, along with two lags of the financial outcome of interest. Once again, the parameters of interest are β_0^h and δ_0^h , where $\{\beta_0^h\}_{h=0}^H$ and $\{\beta_0^h + \delta_0^h\}_{h=0}^H$ are the impulse responses of the financial variables to a monetary policy shock when supply chain pressures are average and tight (one standard deviation above average).

Fig. 5 presents the state-dependent impulse responses. The response of stock prices is largely insignificant when supply chain conditions are average, but there is a highly significant 3%–4% decline in the S&P 500 roughly six to twelve months following a contractionary shock when conditions are tight. This indicates equity values becomes much more sensitive to monetary policy when global supply chain pressures are elevated. This result is complementary to Smirnyagin and Tsyvinski (2022), who find that equity premia increase for firms struck by idiosyncratic supply chain disruptions. The top left panel of Fig. 6 shows the differential response across states (δ_0^h) and confirms that the greater sensitivity is statistically significant.

Next, we turn our attention to the excess bond premium (EBP). The EBP measures the portion of the interest rate spread between an index of corporate bonds and an equivalent maturity government bond that is not due to default risk. It can therefore be interpreted as the difference in credit costs between otherwise equivalent private and government securities that is purely attributable to the existence of financial frictions. The EBP therefore serves as a direct measure of the external finance premium for large firms.¹⁷

The top right panel of Fig. 5 documents a larger response of the EBP when supply chain pressures are elevated. Shortly after the shock, the EBP increases in both supply chain states. However, the increase is more pronounced when supply chain pressures are high. The peak EBP response occurs seven months after the shock. It is 0.09 basis points under average supply chain conditions and 0.22 basis points under tight supply chain conditions – the peak EBP response more than doubles under tight supply chain

¹⁶ The term premia are calculated via the (Kim and Wright, 2005) methodology.

¹⁷ In addition to its appealing conceptual interpretation, Gilchrist and Zakrajšek (2012) also show that the EBP has the strongest forecasting power for economic activity relative to other financial indicators.

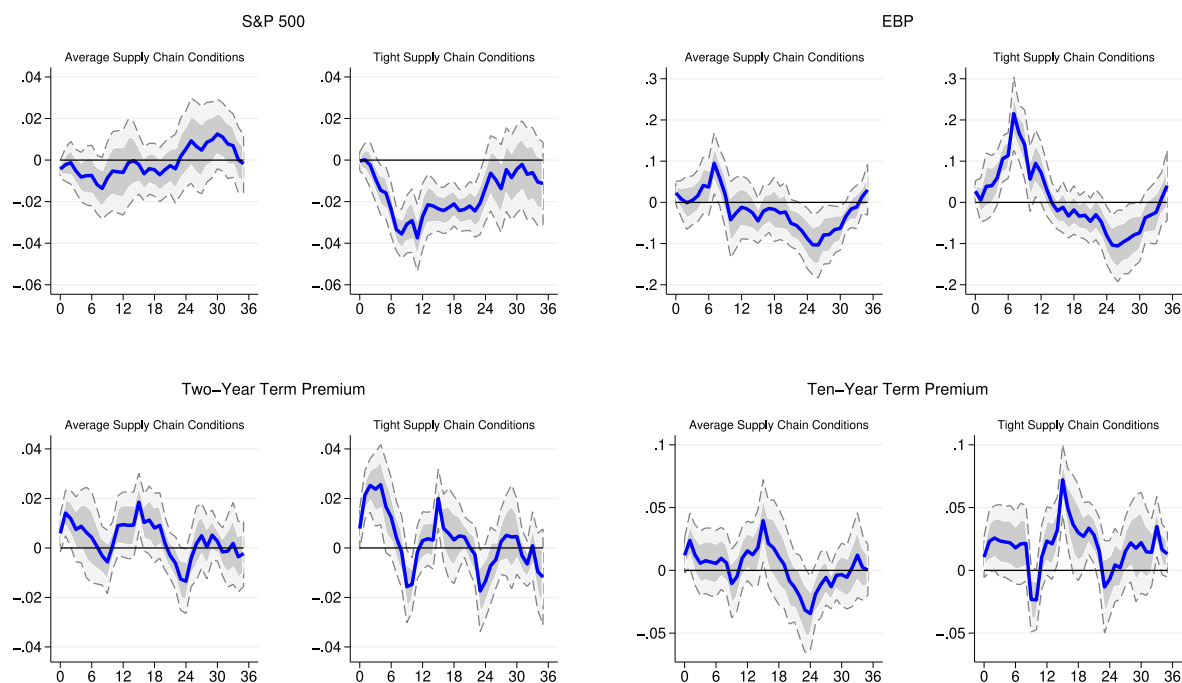


Fig. 5. State-dependent impulse responses. Financial variables. Note: The figure shows the state-dependent impulse response of the S&P 500, excess bond premium, and the two and ten-year term premium to a contractionary monetary policy shock. The left panel shows the response (β_0^h) when supply chain pressures are at their sample (1998–2019) average. The right panel shows the response when supply chain pressures are one standard deviation above average ($\beta_0^h + \delta_0^h$). The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

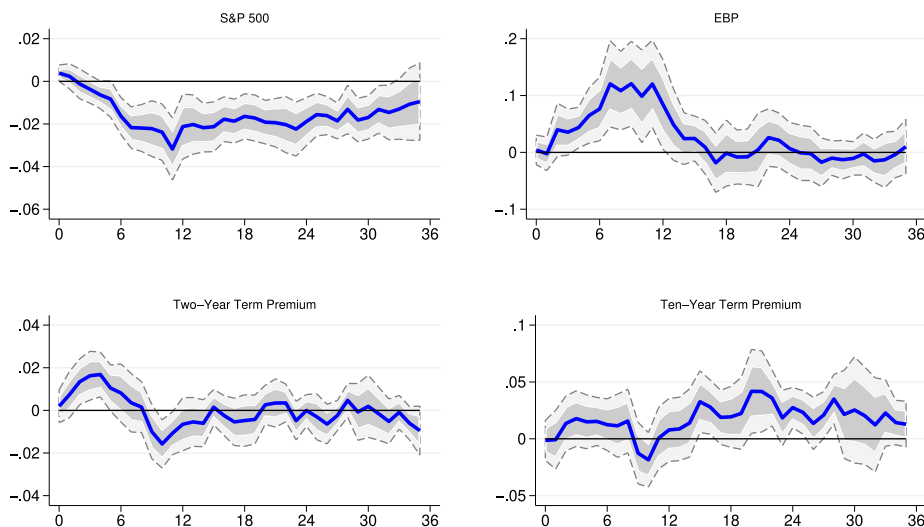


Fig. 6. Differential responses. Financial variables. Note: The figure shows the *additional response* (δ_0^h) of the S&P 500, the excessive bond premium, and the two and ten-year term premium to a contractionary monetary policy shock during a period of elevated supply chain pressure, relative to an average period. The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

conditions. As shown in Fig. 6, the effects across states are also statistically different between four and twelve months after the shock.

Appendix Figure A.3 repeats this exercise for three other measures of credit spreads that do not adjust for default risk. The commercial paper spread responds similarly to the EBP, as there is a significantly larger response around six months after the shock. In addition, the response of the corporate BAA spread is statistically elevated after eighteen months, while the response of the

corporate AAA spread is only statistically amplified after eighteen months under the 68-percent confidence interval.¹⁸ All together, the evidence points towards an increased sensitivity of credit spreads to monetary policy under tight supply chain conditions.

The bottom panels of Fig. 5 show that the responses of Treasury term premia to a contractionary monetary policy shock also vary with the state of supply chain conditions. The two and ten-year premia increase following a monetary tightening, but more so when the shock occurs during tight supply chain conditions. As Fig. 6 shows, the difference for the two-year is only statistically significant between two and six months following the shock. The difference of the ten-year term premia is significant between twenty and twenty-eight months after the shock, but it remains persistently elevated for up to three years after the shock.¹⁹ As term premia represent the extra return that investors require to take on the risk associated with a longer term maturity, their greater responsiveness offers additional evidence that borrowing costs become more sensitive to monetary policy when supply chains are disrupted.

The larger responses of all four financial variables to monetary shocks when the economy faces greater supply chain pressures is consistent with an intensification of the credit channel. A greater sensitivity of the external finance premium when global supply chains are strained rationalizes the amplified effect of monetary policy on key macroeconomic outcomes documented in Section 3.1.

The credit channel implies that financial variables respond to a monetary policy shock more quickly than macroeconomic variables do, as the path of transmission runs from financial factors to macro outcomes. When considering the 36-month impulse responses in Figs. 3–6, it is not clear that the financial variables respond more rapidly than the macro. To provide more precise evidence on the speed of the financial variable response, we turn to high frequency financial market data.

Specifically, we estimate the effect of a monetary policy shock on the daily return on the S&P 500 and daily change in two-year and ten-year Treasury term premia around FOMC announcements. Similar to our local projection specification in Eq. (2), we include a lag of the GSCPI and its interaction with the monetary policy shock.

$$\Delta f_t = \alpha + \beta mp_t + \psi GSCPI_{t-1} + \delta(GSCPI_{t-1} * mp_t) + \epsilon_t \quad (4)$$

We estimate these high frequency regressions over a sample of 167 FOMC decisions occurring between February 1998 and June 2019.²⁰ Following Albagli et al. (2019) and Lakdawala et al. (2021), we focus on a sample excluding the global financial crisis and drop FOMC meetings that took place in 2008 from the estimation. We use the Nakamura and Steinsson (2018) “policy news” shock as our monetary shock measure in this exercise, to avoid endogeneity between the daily change in the two-year yield and the daily change in the two-year term premium. This shock has been normalized so that a unit change is equivalent to a one standard deviation increase in the one year ahead Eurodollar rate. The term premia are once again calculated using the methodology of Kim and Wright (2005). Results are presented in Table 3.

Full sample (excluding 2008) results are shown in Panel (a). Consistent with the existing literature, we find that a surprise tightening of monetary policy results in a significant decline in stock returns and a significant increase in Treasury term premia. The coefficient on the $GSCPI * mp$ interaction term is positive and significant at the 1% level in columns (2) and (3), indicating that the two-year and ten-year premia increase by roughly 37% and 49% more, respectively, following a surprise monetary tightening when global supply chain pressures are one standard deviation above average. While the interaction coefficient in column (1) is insignificant at conventional levels, it is an economically meaningful magnitude, suggesting that the decline in the S&P 500 may be up to 50% larger following a surprise monetary tightening when supply chain pressures are high.

Focusing on a post-GFC (2009–2019) sample in Panel (b) produces more precise stock market results. Across all three columns, we again report the standard result that stock prices decline and term premia rise following a surprise monetary tightening. We also obtain significant interaction coefficients at the 10% level in all three columns, implying that that the standard effects are amplified when global supply chain pressures are elevated in the post-GFC period.

These results around FOMC announcements complement our monthly local projection analysis by providing evidence that (i) financial variables respond immediately to monetary policy shocks, and (ii) the amplification effects documented at a lower frequency in Figs. 5–6 are also visible in the variables’ immediate, high frequency responses.

3.3. Supply chain disruptions amplify the effects of monetary policy - Could other factors drive the results?

Our main results indicate that supply chain disruptions amplify the standard effects of monetary policy. However, is it possible that these findings are contaminated by other confounding factors? For instance, one concern with our empirical approach is that the GSCPI may be contaminated with demand-side factors, in which case the amplified effects of monetary policy in the high GSCPI state may be due to causes unrelated to the state of global supply chains. Benigno et al. (2022) concede that the index may capture some residual demand components during the Global Financial Crisis. To check for the influence of the period around the crisis, we include a Great Recession dummy (taking a value of one from December 2007 to June 2009) and its interactions with $GSCPI$,

¹⁸ The commercial paper, AAA, and BAA spreads are constructed relative to the federal funds rate.

¹⁹ Appendix Figure A.4 repeats this exercise for the one- and five-year Treasury term premia as well. The differential responses are very similar to the two and ten-year, respectively, as there is an immediate and significantly higher spike in the one-year term premium following the shock and a persistently elevated response in the five-year premium.

²⁰ A handful of decisions are excluded due to data availability issues (May 2, 2018) and outlier status (September 17, 2001 and October 8, 2008). The sample begins in February 1998 because that is the first FOMC decision for which GSCPI data becomes available; it ends in June 2019 due to limited availability of some high frequency variables.

Table 3
High frequency responses. Financial variables.

Panel (a): Full sample	(1)	(2)	(3)
	S&P 500	2-Year TP	10-Year TP
mp	−0.416*** [0.139]	0.035*** [0.006]	0.039*** [0.013]
GSCPI	0.026 [0.086]	−0.004* [0.002]	−0.005 [0.004]
GSCPI*mp	−0.208 [0.183]	0.013*** [0.004]	0.019*** [0.007]
Constant	0.304*** [0.084]	−0.003 [0.002]	−0.004 [0.004]
Observations	167	167	167
R-squared	0.071	0.438	0.221
Panel (b): Post-GFC	(1)	(2)	(3)
	S&P 500	2-Year TP	10-Year TP
mp	−0.553*** [0.208]	0.059*** [0.007]	0.090*** [0.020]
GSCPI	0.076 [0.107]	−0.005* [0.003]	−0.009* [0.005]
GSCPI*mp	−0.568* [0.344]	0.014* [0.007]	0.030* [0.017]
Constant	0.234** [0.114]	−0.004 [0.003]	−0.007 [0.007]
Observations	83	83	83
R-squared	0.108	0.615	0.439

Note: The table shows the effect of a monetary policy shock on the daily return on the S&P 500 and daily change in two-year and ten-year Treasury term premia over 167 FOMC announcements from February 1998 to June 2019. Newey–West standard errors are in brackets. $p < 0.10$; $** p < 0.05$; $*** p < 0.01$.

mp , and $GSCPI * mp$, respectively, in Eq. (2). The results in Appendix Figure A.5 show similar amplification effects as the baseline results in Fig. 4.

Furthermore, it may be the case that the heterogeneous impact of monetary shocks documented in our baseline results is driven by underlying factors other than global supply chain conditions. For instance, Tenreyro and Thwaites (2016) show that the response of the US macroeconomy to monetary policy shocks also varies with the state of the business cycle. In our baseline regression, we already include the lags of IP (and other variables) as controls. In Appendix Figure A.6, we go one step further and include the interaction of IP with mp to further control for the state of the business cycle. The baseline results hold through, as δ_0^h is statistically different from zero with a similar magnitude as in Fig. 4.

Recent papers such as Bai et al. (2024) suggest that firm capacity constraints can lead to nonlinear effects of monetary policy. To probe whether this alternative channel could be driving the state dependent effects we include the Federal Reserve Board's Total Capacity Utilization index in our analysis and interact it with the monetary policy shock. Even when controlling for this potential channel, the results in Appendix Figure A.7 continue to show an amplified effect of monetary policy under elevated supply chain pressure.

Additional studies, e.g., Alpanda et al. (2021), examine whether the macroeconomic effects of monetary policy vary with the credit cycle and general financial conditions. In Appendix Figure A.8, we include both the EBP (as before) and its interaction term with mp to control for aggregate financial conditions. As the figure shows, the amplification effects remain valid.²¹ Moreover, in Appendix Figures A.9 and A.10, we add the CBOE Volatility Index (VIX) and Miescu and Mumtaz (2024)'s CE announcement shocks, respectively, along with their interaction terms with mp to Eq. (2) to control for financial stress and for financial shocks. In both cases, the baseline results hold through, as δ_0^h remains statistically different from zero.

Another concern is that the GSCPI may be correlated with omitted global factors, like energy/oil prices or geopolitical risks. This would imply that our estimated amplification effects may not entirely be the result of supply chain pressure but instead might be driven by these other factors. Appendix Figure A.11 shows the estimates of δ_0^h when we include a global energy price index as an additional control variable, along with its interaction with mp . As the figure displays, the baseline results continue to hold, with δ_0^h remaining statistically significant at roughly the same magnitudes as before. Figures A.12 and A.13 show that our main results hold when we add (Känzig, 2021)'s oil surprise and oil news shocks along with their interactions with mp to Eq. (2). In Appendix Figures A.14 and A.15, we add (Caldara and Iacoviello, 2022)'s geopolitical risk index and Ludvigson et al. (2021)'s disaster shocks as well as their interaction terms with mp to the regression, and find very similar amplification effects as in Fig. 4.

To check the robustness to other monetary policy shocks, we employ a second measure akin to Nakamura and Steinsson (2018)'s “policy news shock”, described in Section 2.1. Consistent with the baseline results, Appendix Figure A.16 shows a similarly amplified

²¹ The baseline results, in fact, remain robust to the inclusion of interaction terms of all right-hand side variables from the baseline regression with the monetary policy shock.

effect of the alternative shock measure under elevated supply chain pressures. Compared to Fig. 4, the interaction term becomes statistically significant for a shorter period for IP and the CPI. Another concern in this direction is the contamination of monetary policy shocks via Fed information effects, see e.g., Jarociński and Karadi (2020). To check whether information effects affect our results, we re-estimate Eq. (2) with Bu et al. (2021)'s shock measure which cleanses out potential information effects. The estimates of δ_0^h , shown in Appendix Figure A.17, are once again very similar to our baseline results. Importantly, the δ_0^h estimates for CPI become more strongly significant.

All of the results so far have been based on the GSCPI. One limitation of the GSCPI is that it only becomes available beginning in January 1998. The growing literature has created some alternative indices with earlier start dates. Appendix Figures A.18 and A.19 present the amplification effects when we redo our main exercise with Elsayed et al. (2023)'s GSSDI and with Soto (2023)'s SCBI for our 1998–2019 sample period. The baseline amplification effects remain present using these alternative indices, becoming somewhat stronger for IP and CPI and weaker for retail sales and the unemployment rate. Appendix Figures A.20 and A.21 present differential responses using the alternative indices when the beginning of the sample is extended to January 1992. The responses are similar to those in Figures A.18 and A.19, indicating that our results are robust over the longer sample period.

In Fig. 4, the differential response for CPI is statistically different from zero for only a short duration of the forecast horizon relative to the other variables' responses. To probe the sensitivity of this result, we redo the inflation specification and replace the consumer price index with the personal consumption expenditure index, both with and without food and energy included. Appendix Figure A.22 shows that the amplified impact of monetary shocks on consumer prices does not hinge on the price index used, as the responses of both PCE measures are quite similar to the baseline CPI response. If anything, the δ_0^h coefficients for core PCE suggest a more strongly significant difference after 24 months.

Lastly, we examine potential asymmetries based on the direction of the monetary policy shock. To do so, we construct two new measures: a *contractionary* shock, containing only positive values of mp (capturing unanticipated rate increases), and an *expansionary* shock, containing only negative values of mp (capturing unanticipated rate decreases). We then run a local projection regression that includes separately the contractionary and expansionary shocks as well as their interactions with the GSCPI. Appendix Figure A.23 presents the amplification effects when supply chain conditions are tight for contractionary (left) and expansionary (right) shocks. The results in the left panel are in line with our main results: the effects of contractionary monetary policy shocks are significantly stronger when supply chain conditions are tight. By contrast, we find weaker, if not insignificant, amplification effects for expansionary monetary policy shocks.

These results are consistent with the theoretical reasoning outlined in Section 3.2. When supply chain pressures are elevated, the external finance premium will become more sensitive to *increases* in the policy rate due to both higher borrowing costs and lower expected future cash flows, thereby amplifying the standard effects of a contractionary shock. These two channels may work in opposite directions for an expansionary shock, however. Unanticipated easing by the central bank will reduce the cost of maintaining a larger debt load, thereby generating a larger relative improvement in financial positions and a larger decline in the external finance premium. On the other hand, unanticipated easing may produce *smaller* upward revisions to expected future cash flows in an environment characterized by greater input scarcity and revenue uncertainty. Weaker upward revisions to expected future cash flows generate smaller improvements in financial positions and, consequently, relatively smaller declines in the external finance premium. The overall effect of an expansionary shock is therefore ambiguous, consistent with the weaker responses in Figure A.23.

This section demonstrates that the macroeconomic amplification effects of monetary policy shocks due to supply chain disruptions are unaffected by the inclusion of various different factors, including the business cycle, financial conditions, energy/oil prices, or geopolitical risks. Moreover, our results are robust to different monetary policy shocks, and alternative supply chain and price indices. Consequently, we are confident that our documented amplification effects of monetary policy are not driven by other factors but, in fact, are the result of supply chain disruptions.

4. Firm-level evidence

If an intensification of financial frictions explains the amplified effect of monetary policy on macroeconomic variables during times of high supply chain pressure, we would further expect that (i) firms, on average, have a larger investment response when supply chain conditions are tight and (ii) firms that are more financially constrained experience a larger increase in responsiveness. We provide additional evidence for our proposed mechanism using disaggregated, firm-level data from Compustat. To investigate, we estimate the following panel regression:

$$\Delta \ln(y_{i,t}) = \alpha_i + \beta mp_t + \psi GSCPI_{t-1} + \delta (GSCPI_{t-1} * mp_t) + \Gamma Z_{i,t-1} + \epsilon_{i,t} \quad (5)$$

where $\Delta \ln(y_{i,t})$ is the growth rate of firm i 's net capital stock, α_i is a firm fixed effect, mp_t is our measure of the monetary policy shock, $GSCPI_{t-1}$ is the Global Supply Chain Pressure Index, and $Z_{i,t-1}$ is a vector of firm-level control variables.²² Similar to our local projection specification in Eq. (2), the main objects of interest in Eq. (5) are the coefficients on mp_t and $GSCPI_{t-1} * mp_t$, here denoted as β and δ . They represent the effect of a contractionary monetary policy shock on firm investment when supply chain pressures are normal and the *additional effect* when pressures are elevated. Summary statistics for the firm-level variables used in the estimation are presented in Appendix Table A.1.

²² As is standard, firm controls are included with a lag to mitigate simultaneity concerns.

Table 4
Firm-level evidence.

	(1) All	(2) Young	(3) Old	(4) Small	(5) Large
mp	−0.714*** [0.079]	−0.875*** [0.148]	−0.592*** [0.090]	−0.701*** [0.132]	−0.709*** [0.062]
GSCPI	1.415*** [0.127]	2.091*** [0.261]	1.048*** [0.126]	1.945*** [0.221]	0.759*** [0.107]
GSCPI*mp	−1.564*** [0.131]	−1.843*** [0.252]	−1.358*** [0.143]	−1.697*** [0.213]	−1.385*** [0.116]
Firm controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	190,401	72,924	117,286	104,750	85,407
R-squared	0.110	0.146	0.091	0.117	0.127

Note: The table presents results from estimating Eq. (5). Column (1) reports results for all firms. Columns (2) and (3) split the sample based on the firm age median and report results for firms below and above the median, respectively. Columns (4) and (5) split the sample based on the firm size median and report results for firms below and above the median, respectively. Robust standard errors clustered at the firm-level are in brackets. Firm controls include current to total assets, real sales growth, price-to-cost margin, receivables-minus-payables to sales, depreciation to assets, capitalization, leverage, liquidity, age, and size. Coefficients and standard errors on the control variables are omitted to conserve space (but are available upon request). p < 0.10; ** p < 0.05; *** p < 0.01.

Table 4 shows results from estimating Eq. (5). For the full sample (Column 1), we find that firms decrease their investment following a contractionary monetary policy shock by 0.7 percentage points. The coefficient is statistically significant at the one-percent level. However, when a monetary policy shock hits the economy during a period of elevated supply chain pressure, firms decrease their investment by an *additional* 1.6 percentage points. This coefficient (δ) is statistically significant at the one-percent level. This implies that the average investment response to a monetary shock more than triples when supply chain pressures are a standard deviation above average.

The estimates in Column (1) suggest that investment across all firms becomes more sensitive to monetary policy under tight supply chain conditions. However, previous studies such as Gertler and Gilchrist (1994) and Cloyne et al. (2023) have documented an important dimension of firm heterogeneity, namely, that financially constrained firms have a greater sensitivity to monetary policy than firms that are relatively less constrained. If supply chain pressures exacerbate the effects of monetary policy by intensifying existing financial frictions, we would expect to observe a similar differential sensitivity in the tight supply chain state, i.e., that all firms become more sensitive to a monetary shock, but the increased sensitivity is larger for relatively more constrained firms. To investigate this expected heterogeneity, we require a measure of firm financial constraints. Hadlock and Pierce (2010) show that firm age and size are closely related to the degree to which a firm is financially constrained. Accordingly, we split our sample across these two dimensions and re-estimate Eq. (5).

Columns (2) and (3) present results from dividing the sample of firms into two groups: young and old. Young firms are those below the median age in a given quarter and old firms are those above the median. Both sub-samples decrease investment following a contractionary monetary policy shock under average supply chain conditions. Consistent with the prior literature, the coefficient for young firms is roughly 50 percent larger as for old firms (0.88 vs. 0.60), indicating that financially constrained firms tend to be more sensitive to monetary shocks. Investment decreases *even more* for both groups when supply chain pressures become elevated. Notably, the decline in investment following a contractionary shock is more pronounced for young firms, at 1.84 percentage points, compared to 1.36 percentage points for older firms. This suggests that both types of firms reduce their investment three-times more strongly when supply chains are disrupted, while the additional effect is approximately 40 percent larger for relatively constrained firms.

Columns (4) and (5) present results following a similar sample split based on firm size. Small firms are those below the median level of assets in a given quarter and large firms are those above the median. Once again, investment responds significantly to a monetary policy shock under normal supply chain conditions for both sub-samples, but there is a significantly larger response when the GSCPI is elevated. Small firms experience an additional 1.7 percentage point decline in investment following a contractionary shock, whereas investment at larger firms decreases by an additional 1.38 percentage points, implying that the total effect for small firms is around 15 percent larger relative to large firms. Overall, the results in Table 4 indicate that while heightened supply chain pressures increase the sensitivity of firm-level investment to monetary policy in general, the increase is larger for more financially constrained firms.

To confirm that the greater investment sensitivity for relatively constrained firms is statistically meaningful, we can modify Eq. (5) to incorporate a triple interaction. That is, rather than splitting the sample by age and size, we now interact age/size with mp_t , $GSCPI_{t-1}$, and $GSCPI_{t-1} * mp_t$:

$$\begin{aligned} \Delta \ln(y_{i,t}) = & \alpha_i + \eta_t + \beta (GSCPI_{t-1} * Firm_{i,t-1}) + \psi (mp_t * Firm_{i,t-1}) \\ & + \delta (GSCPI_{t-1} * mp_t * Firm_{i,t-1}) + \Gamma Z_{i,t-1} + \epsilon_{i,t} \end{aligned} \quad (6)$$

The variable of interest in this case becomes the triple interaction $GSCPI_{t-1} * mp_t * Firm_{i,t-1}$, where $Firm_{i,t-1}$ is firm age or size. If the increased sensitivity of investment to monetary policy under elevated supply chain pressures is weaker for less financially

Table 5
Firm-level evidence. Triple interaction.

	(1) Age	(2) Size
mp*Firm	0.014*** [0.005]	0.090** [0.045]
GSCPI*Firm	−0.026*** [0.008]	−0.289*** [0.071]
GSCPI*mp*Firm	0.033*** [0.009]	0.255*** [0.072]
Firm controls	Yes	Yes
Firm fixed effects	Yes	Yes
Time fixed effects	Yes	Yes
Observations	190,401	190,401
R-squared	0.123	0.123

Note: The table presents results from estimating a modified version of Eq. (5). In each column, *GSCPI*, *mp*, and *GSCPI * mp* are interacted with a firm characteristic (*Firm*) that proxies for the degree to which the firm is financially constrained. The firm characteristic in column (1) is age. The firm characteristic in column (2) is size. Robust standard errors clustered at the firm-level are in brackets. Firm controls include current to total assets, real sales growth, price-to-cost margin, receivables-minus-payables to sales, depreciation to assets, capitalization, leverage, liquidity, age, and size. Coefficients and standard errors on the control variables are omitted to conserve space (but are available upon request). $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

constrained (older and larger) firms, we would expect a positive and statistically significant estimate of δ . The results, presented in Table 5, confirm this is indeed the case: the increased investment sensitivity to monetary policy when supply chains are strained is significantly weaker for older and larger firms.

In the estimation up to this point, our right-hand side variables of interest, mp_t and $GSCPI_{t-1}$, have only varied across time. Note that in this specification however, the variable of interest is a triple interaction term containing cross-sectional variation. This makes it possible to include a time fixed effect, which controls for all potentially confounding time-varying macroeconomic phenomena. In addition to the explicit robustness checks performed in Section 3.3, the fact that the estimates in Table 5, which control for all time-varying aggregate conditions, are consistent with the preceding analysis suggests that our local projection results (i.e., Figs. 3–6) are unlikely to have been biased by such confounding factors.

Lastly, we examine potential sectoral heterogeneity. We repeat our main firm-level exercise for the major SIC sectors. Specifically, we estimate Eq. (5) separately for subsamples of firms in each of the ten sectors. If the intensified credit channel is driving our results, we would expect a similar amplification effect across sectors. By contrast, if another factor is driving our results, such as exposure to global supply chains, we would expect to see a greater amplification effect for more exposed sectors, such as manufacturing (LaBelle and Santacreu, 2022). Appendix Table A.2 shows that there is not a clear pattern in the sectors that experience a greater amplification effect, as each industry with a significant negative response to a monetary shock also has a significantly larger response when supply chain pressures are high. Moreover, with the exception of the financial sector, the magnitudes of the significant amplification effects are fairly similar across sectors. The results therefore do not suggest a clear alternative mechanism driving the amplification effect. We interpret this as additional evidence for our proposed credit channel mechanism.

5. Sample extension: COVID-19 pandemic

This Section extends our sample through the COVID-19 pandemic, up to March 2023. The onset of COVID-19 and resulting pandemic led to significant volatility, outliers, and potential structural breaks in many macroeconomic time series, as can be observed for the GSCPI in Fig. 1. Out of an abundance of caution, the focus of our analysis so far has been on the pre-pandemic period where our variables of interest display stable dynamics. Nevertheless, given the dramatic impact of supply chain disruptions on the post-pandemic economy, it remains of great interest to explore whether our conclusion, namely, that supply chain distress amplifies the conventional effects of monetary policy, carries through in the post-pandemic period. Such an extension is particularly important from a policy perspective, as the post-pandemic economy also experienced a sharp pivot in the Federal Reserve's monetary stance.

Fig. 7 shows the responses of industrial production, the CPI, unemployment, and retail sales to a contractionary monetary policy shock over the January 1998 to March 2023 sample period. The figure supports our conclusions from Section 3 – IP, CPI and retail sales decrease following a contractionary monetary policy shock only if the shock hits the economy during elevated supply chain pressure periods. During normal times, the responses are smaller in magnitude and less significant. Appendix Figure A.24 confirms that the amplification effects under tight supply chain conditions are statistically significant after appropriate lags have kicked in.

Comparing the results of Fig. 7 with the pre-pandemic evidence from Fig. 3 suggests that the effects on macroeconomic variables during periods of elevated supply chain pressures for the extended sample period have larger magnitudes (at the trough or peak) and are longer lasting. On the other hand, the lags until the effects become significant are also longer. For instance, the time before the responses of industrial production, inflation and the unemployment rate become significant for the pre-pandemic period is six, eight and two months, respectively. By contrast, it takes way more than one year before the responses become significant when we include the pandemic years in the sample. Finally, for some variables, e.g., the unemployment rate, the initial responses have the opposite sign.

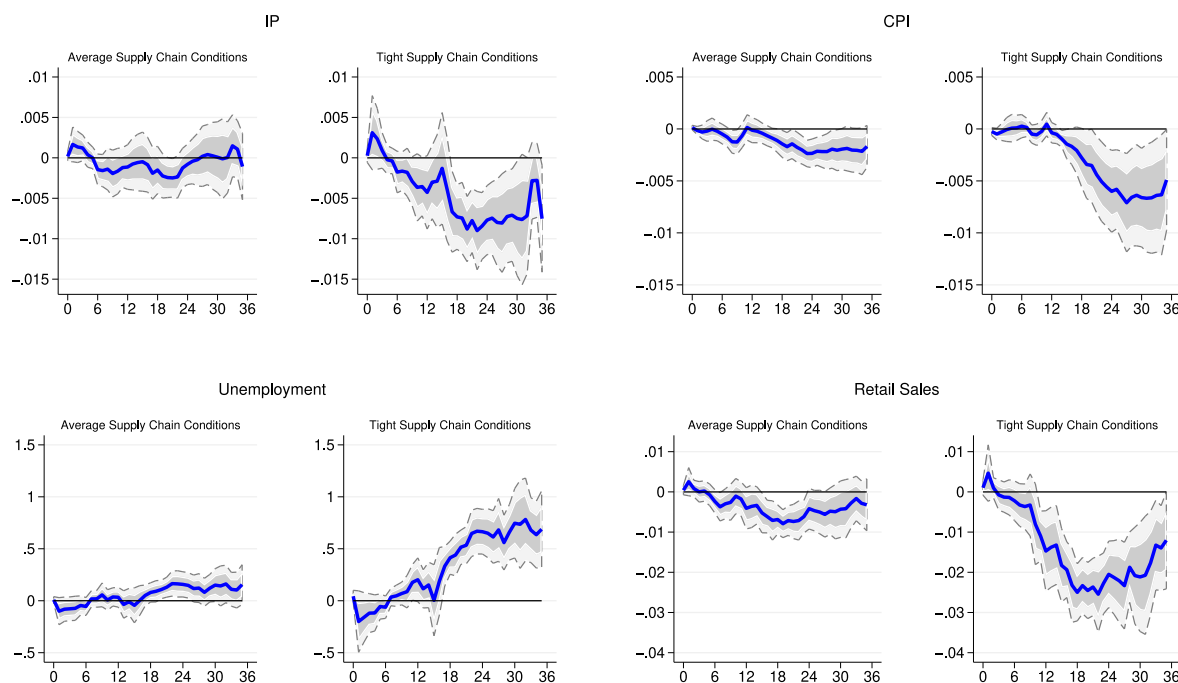


Fig. 7. State-dependent impulse responses. Macro variables. Sample extension. Note: The figure shows the impulse response of industrial production, the consumer price index, the unemployment rate and retail sales to a contractionary monetary policy shock. The sample is extended to run through the COVID-19 pandemic, from January 1998 to March 2023. The left panel shows the response (β_0^h) when supply chain pressures are at their sample average. The right panel shows the response when supply chain pressures are one standard deviation above average ($\beta_0^h + \delta_0^h$). The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

In addition, Fig. 8 displays the responses of the financial variables over the extended sample. The sample once again runs through March 2023, except for the S&P 500, which extends through December 2022 due to data limitations. We once again obtain similar results as Section 3. As with the macro responses, the response of the financial variables during periods of elevated supply chain pressures in Fig. 8 have larger magnitudes at the trough or peak relative to Fig. 5. Appendix Figure A.25 shows that the differences in the responses between normal and elevated supply chain pressure states are statistically significant for stock prices and term premia, while the difference is somewhat more nuanced for the excess bond premium.

Lastly, following Ng (2021), we include national COVID-19 cases and COVID-19 deaths as controls for potentially altered dynamics in the post-pandemic period. We find that the inclusion of COVID-19 cases and deaths as additional controls makes little difference in our estimation. The differential responses of the macro and financial variables, displayed in Appendix Figures A.26 and A.27, are very similar to those shown in Figures A.24 and A.25.

Overall, the conclusions from our baseline analysis hold through, and in some respects become even more striking, when we extend the sample beyond the onset of the COVID-19 pandemic. Supply chain disruptions amplify the conventional impact of monetary shocks on macroeconomic variables as the cost of external finance becomes more sensitive to monetary policy.

6. Conclusion

This paper investigates how global supply chain conditions impact the transmission of monetary policy to the US macroeconomy using a nonlinear local projection framework and the novel GSCPI (Benigno et al., 2022). Our findings reveal that greater pressure on supply chains amplifies the standard effects of monetary policy on industrial production, consumer prices, unemployment and retail sales. We provide evidence that this is due to an intensification of the credit channel: when supply chain pressure is high, the costs of obtaining credit become more sensitive to monetary policy compared to when supply chain pressures are average. Indeed, our results shows that a contractionary monetary policy shock increases the cost of external finance in any state, but significantly more so in periods with greater supply chain pressure.

Of particular interest, given recent policy challenges, is the result that consumer prices are more responsive to monetary policy when supply chain conditions are strained. This could help explain the initial post-pandemic surge in US inflation, when supply chains were very disrupted and the Fed eased policy significantly. The longer estimated monetary policy lags for the extended sample (including the pandemic years) and improved supply chain conditions since 2023 can further help to explain the ongoing persistence of US inflation (the “last mile”) and resilience of the real economy despite the Fed’s 2022–2023 tightening campaign.

Most notably, our results highlight the need for central bank authorities to consider the state of global supply chain conditions when making monetary policy decisions. The post-pandemic literature has developed a number of new measures aimed at

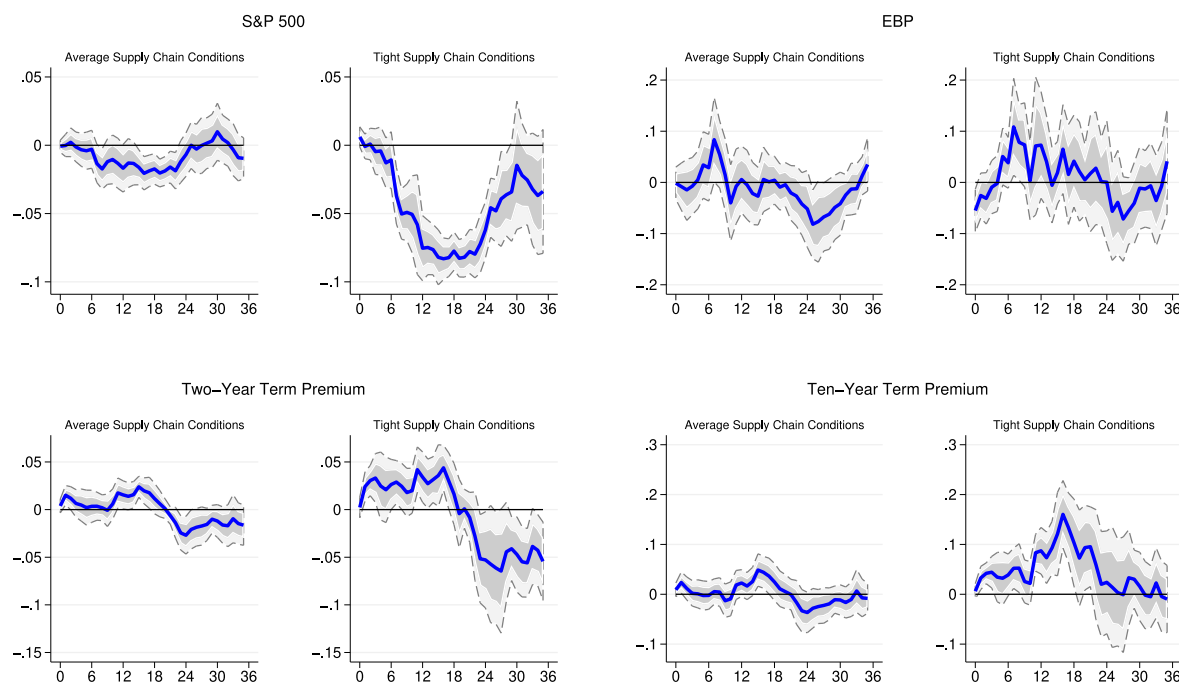


Fig. 8. State-dependent impulse responses. Financial variables. Sample extension. Note: The figure shows the impulse response of the S&P 500, the excess bond premium, and the two and ten-year term premium to a contractionary monetary policy shock. The sample is extended to run through the COVID-19 pandemic, from January 1998 to March 2023. The left panel shows the response (β_0^h) when supply chain pressures are at their sample average. The right panel shows the response when supply chain pressures are one standard deviation above average ($\beta_0^h + \delta_0^h$). The light grey shaded areas indicate 1.65 standard deviation confidence intervals and the dark grey shaded areas indicate 1 standard deviation confidence intervals constructed from Newey–West standard errors.

quantifying the state of supply chain conditions. Indeed, aside from the GSCPI, the GSSDI (Elsayed et al., 2023), the SCBI (Soto, 2023), data on disrupted trade pairs (Smirnyagin and Tsyvinski, 2022), and the Bloomberg Supply Constraint Index all offer various perspectives on supply relationships. All of these sources of information should be monitored and consulted on a regular basis when formulating policy. In addition, more anecdotal information on local supply chain conditions could be compiled in the Beige Book reports prepared for FOMC meetings. Lastly, central bank models can more explicitly incorporate production networks (Acemoglu and Tahbaz-Salehi, 2020; Balleer and Noeller, 2023).

More empirical research on how supply chain disruptions influence macroeconomic outcomes is clearly needed. Such research is particularly relevant as the post-pandemic global economy evolves into new supply chain configurations. As more data becomes available, it will be especially interesting to investigate the impact and persistence of the unprecedented supply chain disruptions associated with COVID-19.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.euroecorev.2024.104949>.

Data availability

Research data is available at this link: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/83K49Y>.

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